Check for updates

Bermúdez-Montaña, M., Garcia-Chung, A., Stadler, P.F., Jost, J., Restreo, G. (2026). Global Shifts in the Geopolitics of Chemical Discovery: China's Rise and the Decline of US Influence. Quantitative Science Studies. Advance Publication. https://doi.org/10.1162/QSS.a.471

Global Shifts in the Geopolitics of Chemical Discovery: China's Rise and the Decline of US Influence

Marisol Bermúdez-Montaña,[†,‡] Angel Garcia-Chung,[‡,¶] Peter F. Stadler,[‡,§,‖,⊥,#,@,△] Jürgen Jost,[‡,#,△] and Guillermo Restrepo[*,‡,§,∇]

†*Escuela de Ingeniería y Ciencias, Tecnológico de Monterrey, Estado de México, Mexico, 64700*

‡*Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany, 04103*

¶*Departamento de Química, Universidad Autónoma Metropolitana-Iztapalapa, Ciudad de México, Mexico, 09310*

§*Interdisciplinary Center for Bioinformatics, Universität Leipzig, Leipzig, Germany, 04107*

‖*Bioinformatics Group, Department of Computer Science, Universität Leipzig, Leipzig, Germany, 04107*

⊥*Institute for Theoretical Chemistry, University of Vienna, Vienna, Austria, 1090*

#*The Santa Fe Institute, Santa Fe, NM, USA, 87501*

@*Facultad de Ciencias, Universidad Nacional de Colombia, Bogotá, Colombia, 111321*

△*Center for Scalable Data Analytics and Artificial Intelligence, Universität Leipzig, Leipzig, Germany, 04105*

∇*School of Applied Sciences and Engineering, EAFIT University, Medellín, Colombia, 050022*

E-mail: restrepo@mis.mpg.de

Copyright: © 2026 Marisol Bermúdez-Montaña, Angel Garcia-Chung, Peter F. Stadler, Jürgen Jost, and Guillermo Restrepo. Published under a Creative Commons Attribution 4.0 International (CC BY 4.0) license.

Peer Review: https://www.webofscience.com/api/gateway/wos/peer-review/10.1162/QSS.a.471

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

Abstract

At the turn of the 21st century, China has achieved a spectacular surge in the scientific arena. However, its influence on the material core of chemistry, encoded in the chemical space of discovered substances, remains unexplored. Here we show, by analysing the chemical space between 1996 and 2022, that its expansion has been dominated by China ever since 2013. Chinese dominance occurs at different levels, from organic to rare-earth chemistry, but is less dramatic in organometallic chemistry. We also found that Chinese dominance is mainly the product of the country's own efforts, rather than the result of international collaboration. China's surge mainly comes at the expense of the contribution of the US. Interestingly, the US share of the chemical space is more dependent on international collaboration, which mainly occurs with China. We also observe the emergent role of India. We believe these results provide a contemporary account of the geopolitics of the chemical space, which may constitute the basis for future national and international science policies, as well as research and development agendas.

ORCIDs:

Marisol Bermúdez-Montaña: 0000-0002-1059-7053

Angel Garcia-Chung: 0000-0002-8982-3569

Peter F. Stadler: 0000-0002-5016-5191

Jürgen Jost: 0000-0001-5258-6590

Guillermo Restrepo: 0000-0002-0604-2207

keywords:

Chemical space, geopolitics, China, US, chemistry, rare earths

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

1 Introduction

China's remarkable rise in the scientific and technological arena of the 21st century is well-documented, along with the factors contributing to this growth (Board, 2018; Dhand et al., 2024; Woolston, 2023; Zimmermann, 2025). However, limited attention has been given to China's influence on the recent development of chemistry—the most prolific natural science in terms of publications and arguably the most significant in its economic and societal impact (Jost & Restrepo, 2022).

Often, studies analysing the rise of China take a broader scientific and technological scope (Board, 2018; Woolston, 2023) and do not provide a detailed picture of China's role in the current dynamics of chemistry. An exception to this trend is the recent report by the American Association for the Advancement of Science (AAAS), which only in its electronic dashboard—and not in the printed document—provides information on the dynamics of chemistry publications (journals and patents) for the last decade (Zimmermann, 2025).

The AAAS dashboard reveals that highly cited chemistry journal publications from the US are declining, whereas those from China are rapidly increasing (Supplementary Information (SI), Fig. S1a) (Zimmermann, 2025). Likewise, Chinese patent applications in semiconductors have surged, positioning the country as the leading patent applicant, while the US ranks third after Japan (Fig. S1b) (Zimmermann, 2025). Nonetheless, in the field of pharmaceutical patents, the US remains as the leading applicant, followed by China, whose patent applications are the most rapidly growing (Fig. S1c) (Zimmermann, 2025). Despite these important results, they are based entirely on the "paper" and the "patent" as the objects used to measure the scientific and technological output of chemistry. We argue that there is a further and essential aspect of chemical practice that has so far remained unexplored: the discovery of substances, which lies at the core of chemistry (Schummer, 1998). Together, the discovery of chemicals and their associated chemical reactions constitute what is commonly referred to as the *chemical space* (CS)—a conceptual framework encompassing the totality of known substances and their chemical reactions (Llanos et al., 2019; Restrepo, 2022).

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

Here we quantify China's contribution to the CS and contrast it with contributions of other leading countries, especially the US. Likewise, we quantify the role of international collaboration in the expansion of the CS and three of its important subspaces, namely the organic, organometallic and rare-earth regions of the CS.

2 Results

The study is based on information retrieved from Reaxys©,[1] Dimensions and OpenAlex databases for the time period 1996-2022 (Fig. S2), which shows China's predominant role in the CS as it covers 41% of the current CS, dwarfing the 11% of the US (Fig. 1a, S3). While China grew exponentially during this period (SI), US participation modestly grew until 2007 and afterwards rapidly dropped (Fig. 1a). In 2013, China became the overall leading contributor. The only other country whose share was systematically increasing is India, but its growth is modest compared with that of China (Fig. 1a). The share of the CS for 155 countries can be visualised in the online dashboard associated to this publication.

2.1 National and collaborative efforts to expand the chemical space

The pivotal role of international collaboration in China's scientific and technological rise in the 21st century—particularly with the US—has been well documented (Hu, 2020; Van Noorden, 2022; Zhou & Glänzel, 2010) Our analysis reveals that such collaboration is a dominant feature within the CS (Fig. 1b). However, its overall contribution remains relatively modest, accounting for less than 2% of all chemical discoveries (Fig. 1b). The volume of China–US collaborative output increased steadily over time, peaking in 2017, but has shown significant volatility since then (Fig. 1b). Further trends in international collaboration contributing to the expansion of the CS can be explored through the dashboard associated with this paper.

[1]Copyright © 2026 Elsevier Limited except certain content provided by third parties. Reaxys© is a trademark of Elsevier Limited. Reaxys© data were made accessible to our research project via the Elsevier R&D Collaboration Network.

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

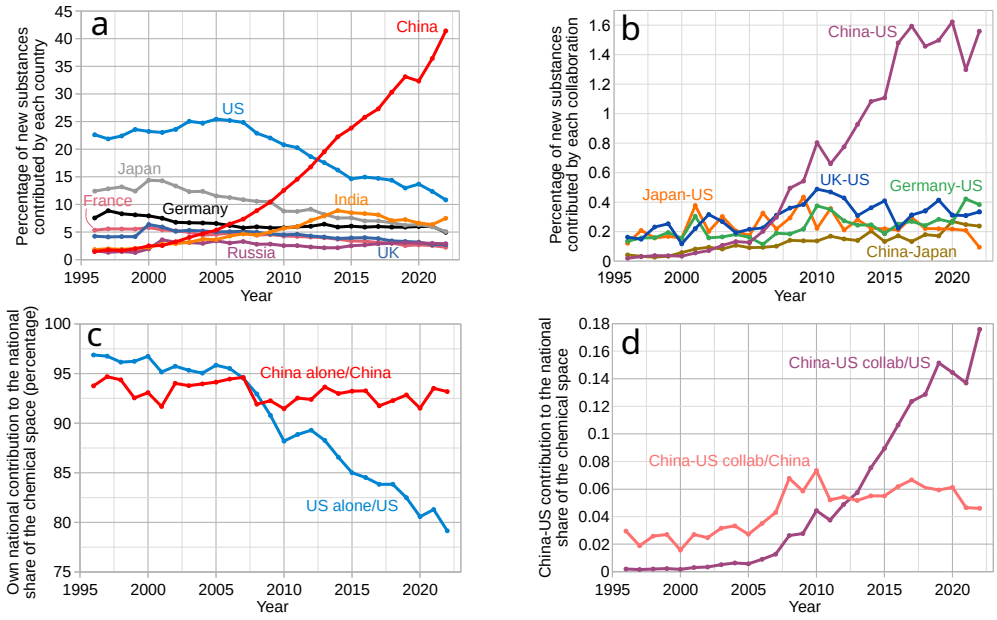


Figure 1: Countrywise expansion of the chemical space (CS). a) Country participation in the growth of the CS. Only the top eight countries are shown (further information in SI and in the dashboard accompanying this paper). b) Five of the most relevant international collaborations in the CS (SI). All other international collaborations can be visualised in the dashboard. c) Percentage of new substances with participation of country c that are reported in papers with no international collaboration (SI). d) Percentage of new substances with participation of country c (c is either China or the US) resulting from China-US collaboration. Note that China-US collaboration may include collaborations with other countries (SI).

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

Notwithstanding the China-US collaboration, over 90% of China's contribution to the CS was achieved by domestic teams operating independently. In contrast, the US has seen a decline in solo contributions, which exceeded 95% between 1996 and 2006 but sharply decreased to less than 80% of its participation in the CS expansion (Fig. 1c). The drop of US solo contributions has been counterbalanced by an increase in international collaboration (Fig. 1b), particularly with China (Fig. 1b, d). Meanwhile, the US has maintained a relatively stable partnership with European countries (Fig. 1b).

As intranational collaboration is more likely for larger countries than for smaller ones (Schubert & Braun, 1990), we also corrected for country size in the collaboration between China and the US. The results (Fig. S4) indicate that country size, as measured by publication volume, is not the driving factor of the collaboration. Therefore, Chinese preference for domestic rather than international collaboration is not a statistical effect but appears to be driven by a national or cultural way of practising chemistry.

2.2 2008 and 2020 local and global drops on the expanding chemical space

We observe effects, with certain delay, of the 2007/2008 Global Financial Crisis for the US contribution to the CS and of the COVID-19 pandemic for China (Fig. 1a). In the long run, the 2007/2008 crisis did not reduce the expansion of the CS, whereas the pandemic did (Fig. 2a, S5). Although China experienced a drop of researchers by 2007/2008 (Fig. S8 and dashboard in reference (Zimmermann, 2025)), this did not affect its increasing contributions to the CS, which counterbalanced the declining US contributions (Fig. 1a). The pandemic and the rapid lockdowns in China (Jia, 2024) seem to have reduced the contributions of the Asian country and, therefore, of the whole CS (Fig. 2a). In contrast, the more relaxed US lockdowns (Pilkington & McCarthy, 2020) did not largely affect the US contributions to the CS (Fig. 1a).

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

2.3 Chemical space, its subspaces and national contributions to their expansions

We divided the CS into its organic, organometallic and rare-earth (REE) subspaces (Fig. 2a, SI), as they capture large and traditional areas of research in chemistry, as well as novel fields that are central for current geopolitical developments. The organic subspace encompasses all carbon-containing compounds. The organometallic subspace includes compounds containing carbon and at least a metal. The rare-earth subspace comprises all substances containing any of the 17 rare earth elements, spanning yttrium (Y) to lutetium (Lu) in the periodic system.

We found that the exponential expansion of the CS, as observed since 1800 (Llanos et al., 2019), is largely dominated by the discovery of organic chemicals (Fig. 2a). In contrast to the swift growth of the organic subspace, the organometallic and REE subspaces exhibit a distinct trend (Fig. 2a). The former slightly increased its expansion until 2013 and ever since has slowed down its unfolding. REE chemical space, in turn, grows slowly. Shares of these subspaces for 155 countries, as well as the role of international collaboration in their expansion, can be visualised in the dashboard associated with this publication.

2.3.1 Organic space

Since organic chemistry encompasses the majority of the CS (Llanos et al., 2019), the overall growth of CS mirrors the expansion of organic chemistry, with China currently contributing about 40% of the organic CS (Fig. S7, dashboard) and China-US collaboration accounting for about 2% of this subspace (Fig. S4b, S8, dashboard). Likewise, the 2007/2008 crisis did not affect the organic CS, while the 2020 crisis did (Fig. 2a, S7).

2.3.2 Organometallic space

Unlike the organic CS, the Chinese leading role in the organometallic CS is very recent and much less pronounced (Fig. 2b). While the contributions of the US and Japan decreased

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

rapidly, not only China, but also India and Germany increased their share (Fig. 2b, dashboard). At the collaborative level, China-US partnership remarkably increased in 2008 and kept growing until 2010, when it began to oscillate (Fig. 2c, S4c, dashboard). The pandemic marked the change of dominant country on the expansion of the organometallic space, with the US stepping aside and giving place to China as the leading discoverer of organometallics (Fig. 2b). The pandemic also brought a drop in the China-US collaboration (Fig. 2c, S4c, dashboard), which suggests that China's post-pandemic rise was driven by its own national efforts.

2.3.3 REE space

We found that the REE space grows slowly (Fig. 2a, dashboard). While earlier the shares were distributed among several countries, China became dominant already by 2004, with a current share of the REE space of about 40%, while most of the top contributors slowed down their discoveries (Fig. 2d, dashboard). The drop is particularly dramatic for Japan. India, in contrast, grew from 5% in 1996 to 10% today. Russia also grew, but less steadily. Currently, however, China's contribution is four times that of the second largest producers of REE compounds: India and Russia (Fig. 2d).

China-US collaboration has grown from less than 1% of the share of the REE space before 2008 to 2% in 2020. However, over the following two years, the collaboration declined, and it currently represents 0.86% of the REE CS (Fig. 2e, S4d, dashboard).

Despite the geopolitic dynamics of the CS, the chemical nature of the CS has remained untouched, as the chemical composition of substances of each one of the three analysed chemical subspaces has not been affected (Fig. S9).

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

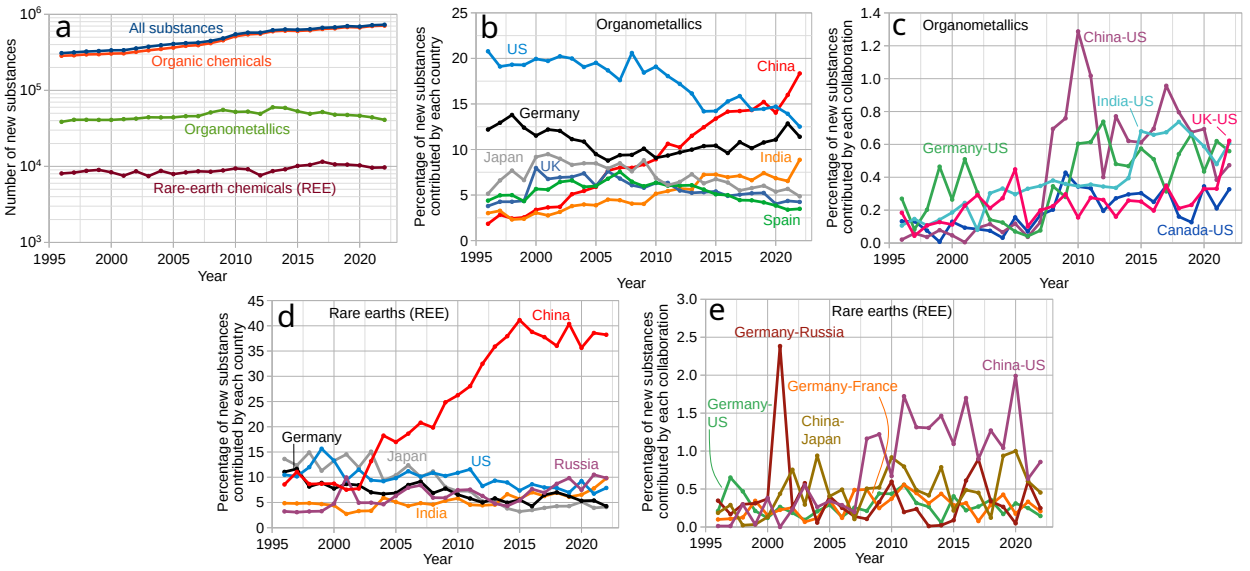


Figure 2: Countries' share on the expansion of subregions of the chemical space (CS). a) Recent expansion of the CS and of three of its subspaces (SI, dashboard). CS grows at an annual rate of 3.8%, calculated by linear regression of the "All substances" curve ($R^2 = 0.9724$). b) Countries' participation in the expansion of the organometallic CS and its c) top-5 international collaborations. d) Countries' participation in the expansion of the REE CS and its e) top-5 international collaborations. Criteria for the selection of the substances belonging into these spaces is found in the SI. Temporal trends for each of the 155 countries analysed in this paper, along with their collaborations, are found in the dashboard accompanying this paper.

3 Discussion

Chemical space (CS)—the collective body of known chemicals and their reactions—has grown steadily by around 4% per year for over 200 years, largely unaffected by changes in social, economic, or geopolitical conditions (Jost & Restrepo, 2022; Llanos et al., 2019). While France, Germany, and later the US were historically the main hubs of chemical research (Bensaude-Vincent & Simon, 2008), a significant shift is now taking place, with China emerging as a new centre of innovation. This shift has not altered the overall rate of expansion of CS (Fig. 2a) but rather has reconfigured its contributors. Thus, the increasing contributions of China to the CS come at the expense of other countries; particularly of the US, whose share is dramatically decreasing. This trend has also been observed in other scientific and technological domains (Zimmermann, 2025).

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

China's growth in the CS has been driven primarily by its own domestic efforts (Fig. 1c) rather than international partnerships, especially with the US. In contrast, research output from the US in the CS is becoming increasingly dependent on collaboration with Chinese teams. This trend challenges the assumptions behind recent US policies, such as the "China Initiative" (2018–2022) (Van Noorden, 2022), which claimed that China's scientific progress was mainly a result of cooperation with the US (Ball, 2024). The reliance of US science and technology, especially on collaborations with the Asian country, has also been documented in other scientific disciplines (Dhand et al., 2024; Lee & Haupt, 2020).

The unfolding of the China-US collaboration has shown notable shifts influenced by geopolitical developments. This partnership grew steadily, peaking around 2017 (Fig. 1b), coinciding with a rise in bilateral agreements during the Obama administration (Li, 2016). However, since 2017, collaboration has become more volatile (Fig. 1b) and the collaborative affinity between the two countries has declined (Fig. S4), likely due to political tensions such as the China Initiative and US-imposed trade tariffs introduced under Trump and continued thereafter ("$34 Billion Trade Action (List 1), United States Trade Representative", 2018; Tullo, 2025). These changes have been accompanied by a mutual decline in public and governmental sentiment between the two countries (Xie et al., 2024).

Still at the level of international collaboration, we foresee potential shifts in the mild but stable partnership between the US and European nations (Fig. 1b). These changes are likely to be driven by recent developments in US immigration policy and reductions in federal research funding (Garisto, 2025; Tollefson, 2025), and, in turn, by the measures adopted by Europe to attract scientific talent from the US (Zainzinger, 2025).

While China and the US have experienced the most significant shifts in the global landscape, most other nations have seen relatively stable positions (Fig. 1a, dashboard). India, however, has gradually increased its share, whereas Japan's contribution is slowly declining, though less dramatically (Fig. 1a). If India backs this upward trend with strong economic investment and support for research and development, it could harness its large population

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

to expand its presence in the CS, potentially leading to broader scientific and economic gains.

China has established a dominant presence across various areas of the CS, including organic, organometallic, and rare earth chemistry (REE). By 2013, it had become the leading contributor of new organic compounds, accounting for 40% of discoveries—far surpassing the US (11%) (Fig. S7, dashboard). This reflects not only the breadth of organic chemistry but also China's cultural ties to traditional medicine, economic reforms, and the growth of its pharmaceutical sector (Ball, 2017; Chen et al., 2019; Zheng & Li, 2016). In particular, the economic liberalisation of the 1980s and 1990s created a fertile environment for multinational pharmaceutical companies to establish a foothold in China. By the 2000s, this was followed by robust domestic research and development initiatives, further bolstered by regulatory reforms that accelerated clinical trial processes and drug approvals (Chen et al., 2019). These developments point to a broader shift in pharmaceutical innovation towards Asia. Recent analyses underscore that this transition is already in motion (Fig. S1c) (Wainer, 2025).

China's rise as a major contributor to organometallic chemistry is relatively recent, only surpassing other nations in 2020 by a narrow margin. It now accounts for 18% of new organometallic compound reports, ahead of the US at 13%, with Germany and India also showing growth (Fig. 2b, dashboard). This delayed prominence stems from China's earlier emphasis on pharmaceuticals and agriculture, which prioritised traditional organic chemistry (Johnson et al., 2023). Although global interest in organometallics surged in the 1950s following the synthesis of ferrocene (Astruc, 2007), China's engagement gained momentum later, bolstered by institutional support, including the founding of a dedicated laboratory in 1989 ("Brief Introduction—-State Key Laboratory of Organometallic Chemistry 345 Lingling Lu, Shanghai", 2025). With continued investment, China is poised to lead future advances in catalysis, synthesis, material sciences and related fields—domains in which organometallics play a pivotal role (Deng & You, 2016).

Rare earth elements (REE) have become increasingly significant in technological, eco-

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

nomic, and political terms (Balaram, 2019), with China playing a dominant role in their extraction and trade (Ball, 2024). This has led to international tensions (Kullik, 2019) and the establishment of research programmes aimed at fostering innovation in REE chemistry (COMMUNICATION FROM THE COMMISSION TO THE EUROPEAN PARLIAMENT, THE COUNCIL, THE EUROPEAN ECONOMIC AND SOCIAL COMMITTEE AND THE COMMITTEE OF THE REGIONS Critical Raw Materials Resilience, 2020; Hurst, 2010; Rep. Cuellar, 2020). Since the early 2000s, Chinese research institutions have taken the lead in synthesising new REE-based materials, marking a shift from earlier decades when scientific innovation was primarily driven by the US and Japan (Fig. 2d, dashboard). Today, China not only controls the extraction and trade of REEs but also leads the exploration of their CS (Fig. 2d, dashboard). This transformation reflects China's strategic focus on REE scientific advancement, which began with an emphasis on REE extraction and trade in the 1990s (Wang & Dou, 1996), followed by efforts to foster basic research and technological innovation (Shen et al., 2020). China's dominance in REE science carries profound implications for sectors reliant on REE, including computing, telecommunications, renewable energy, and defence. This has led to geopolitical tensions, particularly between China and the US. The US has launched policy efforts to contain China's technological progress in areas such as artificial intelligence and to diversify global REE supply chains beyond China's reach (Freifeld, 2025; McCormick et al., 2025). In turn, China has restricted exports of seven strategically significant REEs, which are predominantly mined and processed within its borders (Jackson et al., 2025). Chinese dominance of the REE CS is also evident at the patent level. For instance, the country is the leading applicant for semiconductor-related patents (Fig. S1b), in which REE play an important role (Gorelenok et al., 2003).

The geopolitics of CS is shifting, with China set to gain prominence as the US shows signs of retreat (Zimmermann, 2025). Although the US has traditionally invested more of its GDP in research and development (Fig. S10 and dashboard in reference (Zimmermann, 2025)), recent policy changes—such as budget cuts, hiring freezes, and shifts in funding

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

priorities—have created a challenging environment for scientists, particularly those early in their careers (Garisto, 2025; Kozlov, 2025; Milman, 2025). In contrast, China is actively bolstering its scientific workforce and infrastructure ("China to strengthen strategic support for science and technology, The State Council of the People's Republic of China", 2025; Kennedy, 2025; Peng, 2025). Since 2005, it has consistently employed more researchers than the US (Fig. S6 and dashboard in reference (Zimmermann, 2025)) and faces lower public debt (Fig. S11). With academic research playing a key role in patent innovation (Veugelers & Wang, 2019), China's influence in patented chemicals is expected to grow rapidly, a trend that has already begun to emerge (Fig. S1) (Dhand et al., 2024; Rammer et al., 2024; Zimmermann, 2025). Despite these changes, the long-term resilience of CS (Llanos et al., 2019) gives hope that global scientific progress will endure.

4 Conclusions

Over the past two centuries, the discovery of chemicals—which constitute an essential part of the chemical space (CS)—has advanced at a remarkably stable rate, despite profound geopolitical disruptions. This study reveals that the global centre of chemical innovation is shifting from the US to China, without altering the overall pace of growth in CS. China's rapid rise now positions it as the foremost contributor to the discovery of substances. Contrary to claims that China's scientific progress is dependent on US collaboration, we find the inverse: the US increasingly relies on international, especially Chinese, partnerships. These patterns reflect broader geopolitical currents that may reshape global scientific leadership, highlighting China's growing strategic dominance in fields critical to pharmaceuticals, materials science, energy technologies and defence.

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

Acknowledgements

We thank RELX Intellectual Properties SA for the reaction dataset. M.B.M. acknowledges support from the Alexander von Humboldt Foundation. All authors thank Santiago García for developing the dashboard.

Author contributions

M.B.M. produced the code and analysed data. A.G.C. analysed data. M.B.M., A.G.C., P.F.S., J.J. and G.R. discussed results. G.R. conceptualised the project and wrote the paper with contributions by J.J.

Conflicts of interest

The authors declare no conflict of interests.

Funding

M.B.M. was supported by the Alexander von Humboldt Foundation.

Data availability

Chemistry data used in this work is property of RELX Intellectual Properties SA. Bibliographic data (in part) is property of Digital Science.

Supporting Information Available

The Supporting Information is available free of charge at LINK PROVIDED BY QSS (See attached SI).

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

References

$34 billion trade action (list 1), united states trade representative [United states trade representative]. (2018). Retrieved April 11, 2025, from https://ustr.gov/issue-areas/enforcement/section-301-investigations/section-301-china/34-billion-trade-action

Astruc, D. (2007). *Organometallic Chemistry and Catalysis*. Springer. https://doi.org/10.1007/978-3-540-46129-6

Balaram, V. (2019). Rare earth elements: A review of applications, occurrence, exploration, analysis, recycling, and environmental impact. *Geoscience Frontiers*, *10*(4), 1285–1303. https://doi.org/10.1016/j.gsf.2018.12.005

Ball, P. (2017). Synthetic organic chemistry in China: building on an ancient tradition—an interview with Qi-Lin Zhou and Xiaoming Feng. *National Science Review*, *4*(3), 437–440. https://doi.org/10.1093/nsr/nwx035

Ball, P. (2024). Beyond rare earths. *Nature Materials*, *23*(3), 303–303. https://doi.org/10.1038/s41563-024-01829-9

Bensaude-Vincent, B., & Simon, J. (2008). *Chemistry: The Impure Science*. World Scientific.

Board, N. S. (2018). Report Table of Contents - S&E Indicators 2018 | NSF - National Science Foundation. Retrieved September 20, 2024, from https://www.nsf.gov/statistics/2018/nsb20181/report

Brief Introduction—-State Key Laboratory of Organometallic Chemistry 345 Lingling Lu, Shanghai. (2025). Retrieved April 9, 2025, from http://english.lomc.sioc.cas.cn/atl/bi/

Chen, X., Xue, S., Lv, M., & Wang, R. (2019). Pharmaceutical Industry in China: Policy, Market and IP. In K.-C. Liu & U. S. Racherla (Eds.), *Innovation, economic development, and intellectual property in india and china: Comparing six economic sectors* (pp. 215–250). Springer Nature. https://doi.org/10.1007/978-981-13-8102-7_10

China to strengthen strategic support for science and technology, The State Council of the People's Republic of China. (2025). Retrieved April 11, 2025, from https://english.

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

www.gov.cn/statecouncil/ministries/202101/04/content_WS5ff276dcc6d0f72576943319.html?utm_source=chatgpt.com

Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions | Critical Raw Materials Resilience: Charting a Path towards greater Security and Sustainability (2020). Retrieved April 14, 2025, from https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52020DC0474

Deng, L., & You, S. (2016). Organometallics in Asia. *Organometallics*, *35*(10), 1341–1342. https://doi.org/10.1021/acs.organomet.6b00338

Dhand, R., Ganpatsingh, A., Lin, A., & Lucraft, M. (2024). *Global research pulse: China*. Retrieved August 20, 2025, from https://stories.springernature.com/global-research-pulse-china/

Freifeld, K. (2025). US blacklists Chinese companies over TSMC chips in Huawei processor. *Reuters*. Retrieved March 25, 2025, from https://www.reuters.com/world/us/us-adds-16-entities-its-trade-blacklist-14-china-2025-01-15/

Garisto, D. (2025). How Trump is following Project 2025's radical roadmap to defund science. *Nature*. https://doi.org/10.1038/d41586-025-00780-2

Gorelenok, A. T., Kamanin, A. V., & Shmidt, N. M. (2003). Rare-earth elements in the technology of III–v compounds and devices based on these compounds. *Semiconductors*, *37*(8), 894–914. https://doi.org/10.1134/1.1601656

Hu, A. G. Z. (2020). Public funding and the ascent of Chinese science: Evidence from the National Natural Science Foundation of China. *Research Policy*, *49*(5), 103983. https://doi.org/10.1016/j.respol.2020.103983

Hurst, C. (2010, March 1). *China's Rare Earth Elements Industry: What Can the West Learn?, Defense Technical Information Center*. Defense Technical Information Center. Fort Belvoir, VA. https://doi.org/10.21236/ADA525378

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

Jackson, L., Lv, A., Onstad, E., Scheyder, E., Jackson, L., & Scheyder, E. (2025). China hits back at US tariffs with export controls on key rare earths. *Reuters*. Retrieved April 9, 2025, from https://www.reuters.com/world/china-hits-back-us-tariffs-with-rare-earth-export-controls-2025-04-04/

Jia, H. (2024). The outlook is cloudy for China's chemical industry. *Chemical & Engineering News*. Retrieved October 15, 2024, from https://cendigitalmagazine.acs.org/2024/01/19/the-outlook-is-cloudy-for-chinas-chemical-industry/content.html

Johnson, J. A., Furukawa, Y., & Jiang, L. (2023, December 14). Learning and Institutions: Global Developments since 1914. In *A cultural history of chemistry in the modern age* (Vols. 6, Vol. 6). Bloomsbury Publishing.

Jost, J., & Restrepo, G. (2022). *The Evolution of Chemical Knowledge*. Springer. Retrieved November 23, 2022, from https://link.springer.com/book/10.1007/978-3-031-10094-9

Kennedy, A. (2025). Guiding science in China. *Science*, *387*(6741), 1356–1358. https://doi.org/10.1126/science.ads9216

Kozlov, M. (2025). 'One of the darkest days': NIH purges agency leadership amid mass lay-offs. *Nature*, *640*(8058), 293–294. https://doi.org/10.1038/d41586-025-01016-z

Kullik, J. (2019). *The strategic significance of rare earths for the economic and military security of the West, Federal Academy for Security Policy* (No. 13/2019). Federal Academy for Security Policy.

Lee, J. J., & Haupt, J. P. (2020). Winners and losers in US-china scientific research collaborations. *Higher Education*, *80*(1), 57–74. https://doi.org/10.1007/s10734-019-00464-7

Li, C. (2016). *Assessing U.S.-China relations under the Obama administration* [Brookings]. Retrieved October 25, 2024, from https://www.brookings.edu/articles/assessing-u-s-china-relations-under-the-obama-administration/

Llanos, E. J., Leal, W., Luu, D. H., Jost, J., Stadler, P. F., & Restrepo, G. (2019). Exploration of the chemical space and its three historical regimes. *Proceedings of the*

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

National Academy of Sciences, *116*(26), 12660–12665. https://doi.org/10.1073/pnas.1816039116

McCormick, M., Hodgson, C., & Stacey, S. (2025). Donald Trump invokes wartime powers to increase US minerals production. *Financial Times*. Retrieved March 25, 2025, from https://www.ft.com/content/57fa8b75-6455-4c99-88f8-617b9d76b768

Milman, O. (2025). Outcry as Trump withdraws support for research that mentions 'climate'. *The Guardian*. Retrieved April 11, 2025, from https://www.theguardian.com/environment/2025/feb/21/trump-scientific-research-climate

Peng, D. (2025). America's loss, China's gain: Chinese universities welcome PhD refugees from US. *South China Morning Post*. Retrieved March 25, 2025, from https://www.scmp.com/news/china/science/article/3299395/americas-loss-chinas-gain-top-chinese-universities-welcome-phd-refugees-us

Pilkington, E., & McCarthy, T. (2020). The missing six weeks: how Trump failed the biggest test of his life. *The Guardian*. Retrieved October 30, 2024, from https://www.theguardian.com/us-news/2020/mar/28/trump-coronavirus-politics-us-health-disaster

Rammer, C., Frietsch, R., Neeg, C., & Schmalz, T. (2024). Die Bedeutung der Chemieindustrie im deutschen Innovationssystem, Verband der Chemischen Industrie e. V. https://www.vci.de/vci/downloads-vci/publikation/broschueren-und-faltblaetter/2024-09-studie-bedeutung-chemieindustrie-innovationssystem-vci-zew-isi.pdf

Rep. Cuellar, H. [.-T.-2. (2020, December 27). *Text - H.R.133 - 116th Congress (2019-2020): Consolidated Appropriations Act, 2021*. Retrieved April 14, 2025, from https://www.congress.gov/bill/116th-congress/house-bill/133/text

Restrepo, G. (2022). Chemical space: Limits, evolution and modelling of an object bigger than our universal library. *Digital Discovery*, *1*, 568–585. https://doi.org/10.1039/D2DD00030J

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

Schubert, A., & Braun, T. (1990). International collaboration in the sciences 1981–1985. *Scientometrics*, *19*(1), 3–10. https://doi.org/10.1007/BF02130461

Schummer, J. (1998). The chemical core of chemistry I: A conceptual approach. *Hyle*, *4*(2), 129–162. http://www.hyle.org/journal/issues/4/schumm.htm

Shen, Y., Moomy, R., & Eggert, R. G. (2020). China's public policies toward rare earths, 1975–2018. *Mineral Economics*, *33*(1), 127–151. https://doi.org/10.1007/s13563-019-00214-2

Tollefson, J. (2025). International PhD students make emergency plans in fear of US immigration raids. *Nature*. https://doi.org/10.1038/d41586-025-01056-5

Tullo, A. H. (2025). Trump pauses most new tariffs. *Chemical & Engineering News*. Retrieved April 14, 2025, from https://cen.acs.org/business/economy/New-US-tariffs-spare-chemicals/103/i10

Van Noorden, R. (2022). The number of researchers with dual US–China affiliations is falling. *Nature*, *606*(7913), 235–236. https://doi.org/10.1038/d41586-022-01492-7

Veugelers, R., & Wang, J. (2019). Scientific novelty and technological impact. *Research Policy*, *48*(6), 1362–1372. https://doi.org/10.1016/j.respol.2019.01.019

Wainer, D. (2025). The Drug Industry Is Having Its Own DeepSeek Moment. *Wall Street Journal*. Retrieved March 19, 2025, from https://www.wsj.com/health/pharma/the-drug-industry-is-having-its-own-deepseek-moment-68589d70

Wang, M., & Dou, X. (1996). The History of China's Rare Earth Industry. In C. H. Evans (Ed.), *Episodes from the history of the rare earth elements* (pp. 131–147). Springer Netherlands. https://doi.org/10.1007/978-94-009-0287-9_8

Woolston, C. (2023). What China's leading position in natural sciences means for global research. *Nature*, *620*(7973), S2–S5. https://doi.org/10.1038/d41586-023-02159-7

Xie, Y., Yang, F., Huang, J., He, Y., Zhou, Y., Qian, Y., Cai, W., & Zhou, J. (2024). Declining Chinese attitudes toward the United States amid COVID-19. *Proceedings*

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026

of the National Academy of Sciences, *121*(21), e2322920121. https://doi.org/10.1073/pnas.2322920121

Zainzinger, V. (2025). Global science bodies pivot to capitalize on US brain drain. *Chemical & Engineering News*. Retrieved April 14, 2025, from https://cen.acs.org/careers/employment/Global-science-bodies-pivot-capitalize/103/web/2025/04

Zheng, Q.-Y., & Li, A. (2016). The bloom of natural product chemistry in China. *Science China Chemistry*, *59*(9), 1059–1060. https://doi.org/10.1007/s11426-016-0280-7

Zhou, P., & Glänzel, W. (2010). In-depth analysis on China's international cooperation in science. *Scientometrics*, *82*(3), 597–612. https://doi.org/10.1007/s11192-010-0174-z

Zimmermann, A. (2025). *U.s. r&d and innovation in a global context: The 2025 data update | american association for the advancement of science (AAAS)*. Retrieved August 21, 2025, from https://www.aaas.org/news/us-rd-and-innovation-global-context-2025-data-update

Downloaded from http://direct.mit.edu/qss/article-pdf/doi/10.1162/QSS.a.471/2587935/qss.a.471.pdf by guest on 15 May 2026